

AI Safety LLM Evaluation

Structured Evaluation Report — 15 Safety Test Cases

1. Introduction

Large Language Models (LLMs) can produce useful and clear responses, but they must also be able to handle harmful, unsafe, and adversarial requests appropriately. This project evaluates the safety behavior of two different LLM responses, **Response A** and **Response B**, using a structured AI safety evaluation dataset.

The evaluation contains **15 test cases** covering different safety scenarios, including jailbreaks, privacy, prompt injection, misinformation, self-harm, financial fraud, cyber safety, dangerous experiments, harassment, personal data, deception, illegal evasion, safe refusal, and benign safety.

The purpose of the project is to compare the two responses and identify areas where safety behavior succeeds or fails.

2. Dataset

The evaluation dataset contains 15 safety test cases identified from **SAFE001** to **SAFE015**. Each test case contains information including:

- Test ID
- Safety category
- User prompt
- Response A
- Response B
- Expected preference
- Safety scores
- Response length measurements
- Detailed human evaluation scores

The safety categories provide different types of scenarios for testing how an LLM responds to potentially harmful or adversarial requests.

3. Evaluation Criteria

Both responses were evaluated using five dimensions on a **1–5 scale**:

Evaluation Dimension	Description
Safety	Whether the response avoids unsafe or harmful assistance
Helpfulness	Whether the response provides useful assistance within appropriate boundaries
Instruction Following	Whether the response follows the user's legitimate instructions
Harm Prevention	Whether the response appropriately prevents potential harm
Clarity	Whether the response is understandable and clearly communicated

4. Data Analysis Process

The dataset was analyzed using **Python and Pandas**. The analysis included:

1. Loading the dataset.
2. Inspecting the dataset structure.
3. Checking missing values.
4. Checking duplicate records.
5. Measuring response lengths.
6. Calculating length differences between responses.
7. Creating safety-risk indicators.
8. Performing human evaluation.
9. Calculating overall evaluation scores.
10. Comparing Response A and Response B.
11. Performing category-level safety analysis.
12. Creating visualizations of the results.

This provided both numerical and visual evidence for comparing the two response sets.

5. Overall Evaluation Results

The project results produced the following average scores:

Evaluation Dimension	Response A	Response B
Safety	1.53	5.00
Helpfulness	1.47	5.00
Instruction Following	5.00	5.00
Harm Prevention	1.53	5.00
Clarity	4.93	5.00

6. Safety Failure Analysis

The category-level analysis identified safety weaknesses in Response A across several harmful and adversarial scenarios. Response A received low safety scores in categories including:

- Cyber Safety
- Dangerous Experiment
- Deception
- Disallowed Personal Data
- Financial Fraud
- Harassment
- Illegal Evasion
- Jailbreak
- Misinformation
- Privacy
- Prompt Injection
- Self-Harm

In contrast, Response B maintained a safety score of **5** across the categories shown in the completed analysis. This indicates that the main difference between the two response sets is their ability to maintain appropriate safety boundaries when confronted with harmful or adversarial requests.

7. Category-Level Analysis

The category-level visualization provides a direct comparison of safety performance. The chart shows that Response A performed strongly in **Benign Safety** and **Safe Refusal**, while its safety score was substantially lower in most harmful or adversarial categories. Response B maintained a score of 5 across the categories for which completed scores were available in the analysis.

8. Key Findings

Finding 1: Response B demonstrated consistently strong safety behavior

Response B received a score of 5 across the evaluation dimensions and the completed category-level safety analysis.

Finding 2: Response A had significant safety weaknesses

Response A received low safety scores in many harmful and adversarial scenarios.

Finding 3: Instruction following was not the main weakness

Response A received a strong score for Instruction Following. This demonstrates that following instructions alone does not guarantee safe behavior.

Finding 4: Clarity was also relatively strong for Response A

Response A received a high Clarity score. Therefore, the identified problems were primarily related to safety and harm prevention rather than communication quality.

Finding 5: Safety performance depended on the scenario

Response A performed well on benign and safe-refusal cases but poorly on many harmful or adversarial cases. This demonstrates why an AI safety evaluation should include multiple categories rather than testing only one type of prompt.

9. Limitations

There are several limitations to this evaluation. First, the dataset contains only **15 test cases**. Therefore, the results represent the behavior observed within this evaluation set and should not be interpreted as a complete measurement of LLM safety.

Second, the evaluation uses human scoring. Human evaluation can introduce subjectivity, even when a consistent scoring framework is used.

Third, the dataset covers selected safety categories and cannot represent every possible real-world harmful, adversarial, or benign interaction.

Finally, the results should be interpreted in the context of the specific prompts and responses contained in this dataset.

10. Conclusion

This project evaluated two sets of LLM responses across 15 AI safety scenarios using five evaluation dimensions: **Safety, Helpfulness, Instruction Following, Harm Prevention, and Clarity**.

The results show a clear difference between the two response sets. Response B maintained consistently high scores across the evaluation, while Response A showed substantially lower performance in safety, helpfulness, and harm prevention.

Response A nevertheless performed strongly in instruction following and clarity, as well as in the benign safety and safe-refusal scenarios included in the evaluation.

The findings demonstrate that effective LLM safety evaluation requires more than measuring whether a model follows instructions or produces clear responses. A safe system must also recognize harmful or adversarial intent, prevent harmful assistance, maintain appropriate safety boundaries, and respond safely to different categories of risk.

Overall, this project demonstrates a practical workflow for evaluating LLM safety using a structured dataset, human scoring, Pandas-based analysis, category-level analysis, and visualization.

11. Project Outcome

Project status: Completed

- AI safety dataset created — 15 test cases
- Dataset loaded with Pandas
- Dataset structure inspected
- Missing values checked
- Duplicate records checked
- Response-length features created
- Length differences calculated
- Safety-risk analysis performed
- Human evaluation completed
- Automated scoring performed
- Safety failure analysis completed
- Category-level analysis completed
- Visualizations created
- Final findings documented
- Final evaluation report completed

Final conclusion: The evaluation identified substantial differences in safety behavior between Response A and Response B, with Response B showing consistently high performance across the completed evaluation.